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Spectra II

Last time More on commutative monoids, Spectra, the recognition principle

Today More on Spectra, Stable ∞ -Categories

Recall

> We write $An^{\text{fin}} \subset An$ for the smallest full subcategory containing $\emptyset$ and $*$ and closed under pushouts

A spectrum is a functor $E: An^{\text{fin}}_* \longrightarrow An$ satisfying

(1) $E(*) \xrightarrow{\sim} *$

(2) E carries pushout squares to pullback squares

We write Sp for the ∞ -category of spectra and

$$\begin{aligned} \Omega^\infty: Sp &\longrightarrow An \\ E &\longmapsto E(S^0) \end{aligned}$$

We defined homotopy (abelian) groups $\pi_k(E)$. A spectrum is connective if $\pi_k(E) = 0$ for $k < 0$.

Prop. There is an equivalence of ∞ -categories

$$Sp \xrightarrow{\sim} \lim \left(\cdots \rightarrow An_* \xrightarrow{\Omega} An_* \xrightarrow{\Omega} An_* \right)$$

given on objects by sending a spectrum E to the collection of anima $(E(S^n))_{n \geq 0}$ equipped with the specified equivalences

$$E(S^n) \xrightarrow{\sim} \Omega E(S^{n+1})$$

Recognition principle (Segal). The functor $\Omega^\infty Sp \rightarrow \mathcal{C}Grp(An)$ restricts to an equivalence

$$\Omega^\infty Sp_{\geq 0} \xrightarrow{\sim} \mathcal{C}Grp(An).$$

Digression: the recognition principle for groups

- > There is also a recognition principle for groups. We could have stated it earlier. We won't need it, but it is good to know.

Def. Let $n \geq 0$. An anima X is n -connective if for all $i < n$ and all base points $x \in X$, we have $\pi_i(X, x) \simeq *$.

- > Write $An_{\geq n} \subset An$ for the full subcategory spanned by the n -connective anima.

Recognition principle for groups (Bordmann-Vogt, May [HA, Theorem 5.2.6(10)]).

The functor

$$\Omega : An_* \longrightarrow \text{Grp}(An)$$

restricts to an equivalence $\Omega : An_{\geq 1,*} \xrightarrow{\sim} \text{Grp}(An)$. The inverse is given by

$$B = \text{colim}_{\Delta^{op}} : \text{Grp}(An) \subset \text{Fun}(\Delta^{op}, An) \longrightarrow An.$$

Note that for a monoid M , the colimit is naturally pointed by the 0-Simplices.

> There is a version for $\mathbb{E}_n$ -monoids for all $n \geq 1$, motivated by the following.

Eckmann-Hilton argument Let X be a set equipped with two unital binary operations $(X, \cdot, 1_\cdot)$ and $(X, *, 1_*)$. Assume that for all $a, b, c, d \in X$, we have

$$(a \cdot b) * (c \cdot d) = (a * c) \cdot (b * d).$$

Then $\cdot = *$, $1_\cdot = 1_*$, and this operation is associative and commutative.

Def Let $\mathcal{C}$ be an ∞ -category with finite products and $n \geq 1$. The ∞ -category of $\mathbb{E}_n$ -monoids in $\mathcal{C}$ is

$$\mathrm{Mon}_{\mathbb{E}_n}(\mathcal{C}) := \underbrace{\mathrm{Mon}(\mathrm{Mon}(\dots \mathrm{Mon}(\mathcal{C})))}_{n\text{-times}}$$

We have forgetful functors

$$\text{Mon}_{\mathbb{E}_{n+1}}(e) \longrightarrow \text{Mon}_{\mathbb{E}_n}(e),$$

We say that an $\mathbb{E}_n$ -monoid is an $\mathbb{E}_n$ -group if the underlying $\mathbb{E}_1$ -monoid is a group.

Ex. For any pointed object $x \cdot 1e \rightarrow X$, the n -fold loop object is naturally an $\mathbb{E}_n$ -group.

Recognition principle for $\mathbb{E}_n$ -groups. The functor

$$\Omega^n: \text{An}_* \rightarrow \text{Grp}_{\mathbb{E}_n}(\text{An})$$

restricts to an equivalence $\Omega^n: \text{An}_{\geq n, *} \xrightarrow{\sim} \text{Grp}_{\mathbb{E}_n}(\text{An})$. The inverse is given by applying B n times.

Prop. There is an equivalence of ∞ -categories

$$\text{CMon}(\text{An}) \xrightarrow{\sim} \lim \left(\cdots \longrightarrow \text{Mon}_{\mathbb{E}_{n+1}}(\text{An}) \longrightarrow \text{Mon}_{\mathbb{E}_n}(\text{An}) \longrightarrow \cdots \right)$$

that restricts to an equivalence

$$\mathcal{C}\mathrm{Grp}(A_n) \xrightarrow{\sim} \lim(\cdots \rightarrow \mathrm{Grp}_{\mathbb{E}_{n+1}}(A_n) \rightarrow \mathrm{Grp}_{\mathbb{E}_n}(A_n) \rightarrow \cdots)$$

Excisive approximation

Def. Let $\mathcal{C}$ be an ∞ -category. An object $0 \in \mathcal{C}$ is a zero object if 0 is both initial and terminal. We say that $\mathcal{C}$ is pointed if $\mathcal{C}$ admits a zero object.

If $\mathcal{C}$ is a pointed ∞ -category and $f: X \rightarrow Y$ is a map in $\mathcal{C}$, the fiber of f is the pullback (if it exists)

$$\begin{array}{ccc} \text{fib}(f) & \longrightarrow & X \\ \downarrow & \lrcorner & \downarrow f \\ 0 & \longrightarrow & Y \end{array}$$

The cofiber of f is the pushout

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow & \lrcorner & \downarrow \\ 0 & \longrightarrow & \text{cofib}(f) \end{array}$$

A commutative square

(*)

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow & & \downarrow g \\ 0 & \longrightarrow & Z \end{array}$$



This square doesn't just say there exists an equivalence $gf \simeq 0$, but provides the data of one.

is a fiber sequence if it is a pullback, a cofiber sequence if it is a pushout, and an exact sequence or bifiber sequence if it is both

> If $\mathcal{C}$ is pointed, a functor $F: \mathcal{C} \rightarrow \mathcal{D}$ is reduced if F sends the zero object to the terminal object of $\mathcal{D}$. We write $\text{Fun}_*(\mathcal{C}, \mathcal{D}) \subset \text{Fun}(\mathcal{C}, \mathcal{D})$

for the full subcategory spanned by the reduced functors

Lem. The inclusion $\text{Fun}_*(\text{An}_*^{\text{fin}}, \text{An}) \hookrightarrow \text{Fun}(\text{An}_*^{\text{fin}}, \text{An})$ admits a left adjoint $(-)^{\text{red}}$ given by the formula

$$F^{\text{red}} := \text{Cofib}(F(*) \longrightarrow F)$$

Here, the left-hand side denotes the constant functor with value $F(*)$ and the natural transformation comes from the fact that $*$ is initial in An_*^{fin} .

Obs. If $F: \text{An}_*^{\text{fin}} \rightarrow \text{An}$ is reduced, for each $X \in \text{An}_*^{\text{fin}}$, the universal properties of the pushout and pullback induce a map

$$\eta_X: F(X) \longrightarrow \Omega F(\Sigma X)$$

apply F to

$$\begin{array}{ccc} X & \longrightarrow & * \\ \downarrow & \lrcorner & \downarrow \\ * & \longrightarrow & \Sigma X \end{array}$$

natural in X

Lem. The inclusion $\text{Sp} \hookrightarrow \text{Fun}_*(\text{An}_*^{\text{fin}}, \text{An})$ admits a left adjoint P_1 given by the formula

$$P_1(F) := \text{colim}(F \xrightarrow{\eta} \Omega F \Sigma \xrightarrow{\eta} \Omega^2 F \Sigma^2 \rightarrow \dots)$$

We call $P_1(F)$ the 1st excisive approximation of F .

Cor The inclusion $Sp \hookrightarrow \text{Fun}(An_*^{\text{fin}}, An)$ admits a left adjoint p_1^{red} . In particular, Sp has all limits and colimits.

Suspension Spectra & the Sphere

Def. Let $\mathcal{C}$ be a pointed ∞ -category with finite products and pushouts. Observe that for all $X, Y \in \mathcal{C}$, the collapse maps $X \vee Y \rightarrow X$ and $X \vee Y \rightarrow Y$ induce a natural map

$$\underbrace{\quad}_{\text{coproduct}} \quad X \vee Y \rightarrow X \times Y$$

The Smash product $X \wedge Y$ is the cofiber

$$X \wedge Y = \text{cofib}(X \vee Y \rightarrow X \times Y)$$

Observe we have $X \wedge 0 \simeq 0$.

Def. Let $X \in \mathcal{A}n_*$ be a pointed anima. Note that by the above, $X \wedge (-) : \mathcal{A}n_*^{\text{fin}} \rightarrow \mathcal{A}n$ is reduced. The Suspension Spectrum of X is the 1st excisive approximation

$$\Sigma^\infty X := P_1(X \wedge (-)).$$

The assignment $X \mapsto \Sigma^\infty X$ defines a functor $\Sigma^\infty \mathcal{A}n_* \rightarrow \mathcal{S}p$.

> The following comes from unwinding what we've said so far.

Lem. The functor $\Sigma^\infty: \mathbf{An}_* \longrightarrow \mathbf{Sp}$ is left adjoint to $\Omega^\infty: \mathbf{Sp} \longrightarrow \mathbf{An}_*$.

Obs. If $\mathcal{C}$ is an ∞ -category with finite coproducts and a terminal object, then the forgetful functor

$$\mathcal{C}_* \longrightarrow \mathcal{C}$$

admits a left adjoint $(-)_+: \mathcal{C} \longrightarrow \mathcal{C}_*$ defined by

$$X_+ = X \sqcup 1_{\mathcal{C}}$$

Cor. The functor $\Omega^\infty: \mathbf{Sp} \longrightarrow \mathbf{An}$ admits a left adjoint $\Sigma_+^\infty: \mathbf{An} \longrightarrow \mathbf{Sp}$ defined by

$$\Sigma_+^\infty(X) = \Sigma^\infty(X_+)$$

Def. The sphere spectrum is the suspension spectrum

$$\mathbb{S} := \Sigma^\infty S^0.$$

Lem. The suspension spectrum functor $\Sigma^\infty : \mathcal{A}n_* \rightarrow \mathcal{A}n$ factors through $\mathcal{S}p_{\geq 0}$.

Lem. Under the identification $\Omega^\infty : \mathcal{S}p_{\geq 0} \xrightarrow{\sim} \mathcal{C}Grp(\mathcal{A}n)$, the sphere spectrum corresponds to the group completion $(\text{Set}^{\text{fin}}, \wedge)^{gp}$.

Stability

Observation We can also describe $\mathcal{S}p$ as the limit of the $\mathbb{Z}$ -indexed diagram

$$\xrightarrow{\Omega} An_* \xrightarrow{\Omega} An_* \xrightarrow{\Omega} An_* \xrightarrow{\Omega} \dots$$

Then a spectrum is a sequence of pointed anima $(E_n)_{n \in \mathbb{Z}}$ together with equivalences $(E_n \xrightarrow{\sim} \Omega E_{n+1})$. Under this identification, it is easy to check that the loops functor $\Omega: \mathcal{S}p \rightarrow \mathcal{S}p$ corresponds to the auto-equivalence

$$(E_n, E_n \xrightarrow{\sim} \Omega E_{n+1})_{n \in \mathbb{Z}} \mapsto (E_{n-1}, E_{n-1} \xrightarrow{\sim} \Omega E_n)_{n \in \mathbb{Z}}.$$

Def. A pointed ∞ -category $\mathcal{C}$ is **stable** if every map admits a fiber and $\Omega: \mathcal{C} \rightarrow \mathcal{C}$ is an equivalence.

Characterization of Stable ∞ -categories. Let $\mathcal{C}$ be a pointed ∞ -category. TFAE:

(1) $\mathcal{C}$ is stable.

(2) Every map in $\mathcal{C}$ admits a cofiber and $\Sigma: \mathcal{C} \rightarrow \mathcal{C}$ is an equivalence.

(3) Every map in $\mathcal{C}$ admits a fiber and a cofiber and a null sequence

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow & & \downarrow g \\ 0 & \longrightarrow & Z \end{array}$$

is a fiber sequence iff it is a cofiber sequence.

(4) $\mathcal{C}$ has pushouts and pullbacks and a commutative square

$$\begin{array}{ccc} W & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y & \longrightarrow & Z \end{array}$$

is a pushout if and only if it is a pullback.

(5) $\mathcal{C}$ has finite limits and finite colimits, and they commute with each other

In this case, $\Sigma: \mathcal{C} \rightarrow \mathcal{C}$ and $\Omega: \mathcal{C} \rightarrow \mathcal{C}$ are inverse equivalences of ∞ -categories.

Ex

(0) The category with one obj and only the identity map is stable

(1) The ∞ -category of Spectra is stable

(2) If A is a ring, the derived ∞ -category $D(A)$ is stable.

(3) If $\mathcal{C}$ is a stable ∞ -category and I is any ∞ -category, then $\text{Fun}(I, \mathcal{C})$ is stable.

Exercise. If $\mathcal{C}$ is a nontrivial 1-category, then $\mathcal{C}$ is not stable.

Ex. The ∞ -category $\text{Grp}(An) \simeq \text{Sp}_{\geq 0}$ is not stable

Ex. An_* is not stable

Relationship to triangulated categories

- > Let's explain why stability is the natural ∞ -categorical enhancement of the theory of triangulated categories
 - One big advantage is that stability is a property of an ∞ -category, whereas one has to put a triangulated structure on a 1-category.

Ntn If $\mathcal{C}$ is a stable ∞ -category, we write $\Sigma^{-1} = \Omega$. For each $n \in \mathbb{Z}$, we write $(-)[n] := \Sigma^n$. We call $(-)[1] \mathcal{C} \xrightarrow{\sim} \mathcal{C}$ the shift functor.

Axiom Let $\text{Cat}_1 \subset \text{Cat}_\infty$ denote the full subcategory spanned by the 1-categories, i.e., those ∞ -categories whose mapping anima are sets.

This inclusion admits a left adjoint $h = h_1: \text{Cat}_\infty \rightarrow \text{Cat}_1$ called the homotopy (1-)category. Moreover, h preserves finite products, and the unit $\mathcal{C} \rightarrow h\mathcal{C}$ is full and essentially surjective on objects.

Prop [HA, Theorem 1.12.14] Let $\mathcal{C}$ be a stable ∞ -category. Then the homotopy category $h\mathcal{C}$ has a natural triangulated structure with:

(1) Shift induced by $(-)[1]: \mathcal{C} \xrightarrow{\sim} \mathcal{C}$

(2) A diagram

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$$

in $h\mathcal{C}$ is a distinguished triangle if there is a commutative diagram in $\mathcal{C}$

$$\begin{array}{ccccc} X & \xrightarrow{\tilde{f}} & Y & \longrightarrow & 0 \\ \downarrow & \square & \downarrow \tilde{g} & \square & \downarrow \\ 0 & \longrightarrow & Z & \xrightarrow{\tilde{h}} & W \end{array}$$

where both squares are pushouts, $\tilde{f}$ and $\tilde{g}$ have images f and g in $h\mathcal{C}$, and the map h is the composite of the class of $\tilde{h}$ with the equiv. $W \simeq X[1]$ determined by the large rectangle.

The universal property of Spectra

Thm (universal property of Spectra), Let $\mathcal{C}$ be a stable ∞ -category with colimits. Then evaluation on the sphere spectrum defines an equivalence of ∞ -categories

$$\begin{array}{ccc} \mathrm{Fun}^{\mathrm{colim}}(\mathrm{Sp}, \mathcal{C}) & \xrightarrow{\sim} & \mathcal{C} \\ F & \longmapsto & F(S) \end{array}$$

Moreover, every colimit-preserving functor $\mathrm{Sp} \rightarrow \mathcal{C}$ is a left adjoint.

> There is a version without assuming colimits exist.

Def A functor $F: \mathcal{C} \rightarrow \mathcal{D}$ between stable ∞ -categories is exact if F preserves the zero object, pushouts, and pullbacks.

Def The full subcategory $\mathrm{Sp}^{\mathrm{fin}} \subset \mathrm{Sp}$ of finite spectra is the smallest full subcategory containing 0 and S and closed under pullbacks and pushouts. Note that $\mathrm{Sp}^{\mathrm{fin}}$ is stable.

Thm (universal property of finite spectra). Let $\mathcal{C}$ be a stable ∞ -category. Then evaluation on the sphere spectrum defines an equivalence of ∞ -categories

$$\begin{array}{ccc} \mathrm{Fun}^{\mathrm{ex}}(\mathrm{Sp}^{\mathrm{fin}}, \mathcal{C}) & \xrightarrow{\sim} & \mathcal{C} \\ F & \longmapsto & F(S) \end{array}$$